

Individuality, Boundary Cost, and Cohesion

Professional reading edition of the evolving public MKUFT canon

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Purpose

This document explores one structural idea:

A persistent individual system must maintain distinctions between itself and its surroundings, and maintaining those distinctions carries cost.

The idea is useful across several domains, but the mechanisms are not identical. A cell membrane, a cognitive identity, a political border, and a physical field boundary cannot be treated as one mechanism merely because they share boundary language.

1. Inside, outside, and persistence

A system becomes identifiable when some relation distinguishes inside from outside, self from environment, or one state from another. That distinction may be physical, biological, informational, cognitive, social, or procedural.

Persistence costs may include physical energy, material replacement, error correction, regulatory effort, contradiction, or information loss. Outside physics, “cost” does not automatically mean physical energy and must be operationalised for the active domain.

2. Persistence and equilibration

Many systems operate between two failure tendencies: too little boundary or control allows distinction to dissolve, while excessive rigidity reduces adaptation and may increase fracture risk.

This is a general systems pattern rather than a universal law that every stable system literally oscillates in the same manner.

3. Dimensionality hypothesis

In n spatial dimensions, a codimension-one boundary may enclose an n -dimensional region where the relevant topology permits it. Three-dimensional space specifically permits familiar two-dimensional closed surfaces enclosing volume, together with the rotational and orientational structure observed in ordinary physical systems.

MKUFT treats the usefulness of three dimensions for persistent bounded physical structures as a question worth examining.

A stronger claim—that three dimensions are uniquely cheapest for individuality—would require a declared class of comparable systems in other dimensions; defined boundary, stability, and maintenance costs; matched topology and dynamics; proof or numerical comparison; and evidence that the result is not an artefact of the chosen model.

Until such a comparison exists, the claim remains heuristic.

Higher-dimensional state spaces used to represent information, cognition, or control variables are not additional physical spatial dimensions. Traversal through those spaces is not literal physical movement without an independent physical model.

4. Boundary, gradient, and response

A maintained boundary can support a difference across it. In physical systems, a maintained difference may create a gradient, and a gradient may drive transport, force, flux, or other domain-specific behaviour according to established equations.

The clean structural sequence is **boundary → maintained difference → gradient → domain-specific response**.

The final response must not automatically be renamed a physical field in cognitive, social, or symbolic domains.

5. Physical scale

At the physical layer, claims require equations, units, measurements, and ordinary physical baselines.

Mass, charge, spin, gravity, and electromagnetism already have precise scientific meanings. MKUFT may compare them through constraint and persistence language, but it cannot redefine them as “leakage” or “boundary asymmetry” without deriving accepted limits and making distinct predictions.

This document is therefore a conceptual bridge rather than a replacement microphysical theory.

6. Biological scale

Cells provide a concrete example of maintained individuality through membrane integrity, regulated exchange, metabolism, signalling, and repair.

A useful structural summary is **membrane integrity + regulated exchange + energy use + error correction → persistent living organisation**.

Biological death cannot always be reduced to one boundary failure, but irreversible loss of integrated regulation is a relevant systems comparison.

7. Cognitive scale

At the cognitive layer, boundary language may refer to self-model stability, attention allocation, memory integration, threat gating, openness to evidence, and protection against fragmentation.

Trauma, dissociation, and healing are clinically complex and cannot be reduced to one formula. MKUFT uses boundary and accessibility language as a systems aid subordinate to psychological and medical evidence.

8. Social scale

Social systems maintain roles, norms, institutions, trust, enforcement, resources, and borders.

One useful systems question is:

Does the system spend more on maintaining contradiction and coercion than it gains from coordination?

Trust can reduce transaction and enforcement costs. Coercion can temporarily stabilise a system while increasing long-term fragility. These are social-system claims rather than evidence that love or cohesion is a physical force.

9. Cosmic scale

Galaxies, gravitational structure, expansion, and dark matter require astrophysical evidence and quantitative comparison with established models.

MKUFT may ask whether hidden constraints can be represented alongside missing-mass hypotheses, but it does not currently solve dark matter, gravitation, or cosmic expansion.

10. Love as low-cost human cohesion

In human and social systems, love, trust, care, truthful relation, and mutual recognition can reduce coordination cost while preserving individuality. A fear-based system may require surveillance, threat, concealment, and repeated enforcement, whereas deserved trust can carry complexity with less friction when boundaries remain intact.

This is a psychological and social systems proposition, not a new physical interaction.

Within MKUFT's metaphysical interpretation, Love is treated more strongly as a load-bearing unity-principle. Measurable social proxies neither exhaust nor prove that metaphysical claim.

11. Boundaries and intelligence

Novel information often appears at interfaces where systems encounter difference: sensory boundaries, ecological edges, disciplinary crossings, social frontiers, and changing task conditions.

The narrower hypothesis is:

Managing meaningful difference can create pressure for sensing, modelling, prediction, and adaptive control.

That does not imply that every boundary produces intelligence.

12. Traversal cost

For a path γ within one declared state space, a candidate path cost is

$$C[\gamma] = \int_{\gamma} \lambda(x) ds,$$

where $\lambda(x)$, ds , and the resulting units must be defined for the domain.

A Gibbs-like path model requires a dimensionless exponent. One form is

$$P(B | A) = \frac{1}{Z_A} \sum_{\gamma \in \Gamma(A \rightarrow B)} \exp[-\beta C[\gamma]],$$

where β has reciprocal units of C or follows an explicit normalisation convention.

These expressions test whether lower-cost and more numerous admissible routes improve prediction. They are not universal laws, and an abstract route is not automatically a physical trajectory.

13. Falsifiers and limits

The bridge is weakened if boundary language adds no explanatory or predictive value; one domain's evidence is used as proof in another; physical terms are applied metaphorically without distinction; dimensional claims lack an explicit comparison model; state-space dimension is confused with physical spatial dimension; social or psychological claims perform worse than established theories; or “cost” cannot be defined for the active domain.

14. Related public documents

- [Standalone Formal Addendum](#)
- [Cross-Layer Invariants and Layer Addressing](#)
- [Typed Traversal and Equation Hygiene](#)
- [Agency Accessibility and Capture Geometry](#)
- [Physics-Facing MKUFT Explanation](#)
- [Falsification Summary](#)
- [Cross-Support and Traversal Map](#)

Summary

Persistent individuality requires maintained distinction, and maintained distinction usually carries cost. The useful questions are: what boundary is maintained, what difference does it preserve, what does maintenance cost, what happens when the boundary becomes too weak or too rigid, and which dimensions are physical rather than abstract or analogical?

The mechanism must be established separately at each layer.